Practical Fairness: An Evaluation of Bias Mitigation Strategies for NLP Applications

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ABSTRACT

As the use of algorithmic decision-making becomes more widely adopted, there is a growing need for practical strategies to mitigate bias in automated systems, particularly in natural language processing. Due to a lack of standardization, comprehensive evaluation metrics, and real-world evaluation, comparing different mitigation strategies is challenging. In this work, we demonstrate the presence of gender bias when using the large language model (LLM) RoBERTa [9] to classify occupation from text samples similar to LinkedIn bios. We then provide a holistic evaluation of ten bias mitigation strategies applied to a common problem, providing insights into their effectiveness and trade-offs under different constraints. Based on this analysis, we offer guidance for companies on selecting the most suitable strategy for a given circumstance with consideration to computational constraints, metric priorities, and dataset characteristics. This work contributes a consistent assessment of the efficacy and trade-offs of common bias mitigation techniques that could be used in real-world applications as well as a point of comparison for future methodologies.

KEYWORDS

Bias, Natural Language Processing, Fairness, Text Classification, Bias Evaluation, Bias Mitigation, Gender Bias

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1 INTRODUCTION

Recent developments in natural language processing have led to an unprecedented number of institutions promising to integrate AI into their workflows. The surge of public commercial attention has brought heightened awareness to the potential harms of biased AI, and recent policy emphasizes that bias will need to be addressed for future deployments. While much of the attention focuses on generative applications, many companies will realistically begin to integrate AI into their workflows by using smaller scale classifiers to automate decisions. If these decisions are subject to bias, such as who to hire and to whom to grant loans, they can have major ethical impacts on society. Gender bias poses a particular problem for natural language processing because the gender of a subject is explicitly and implicitly represented in most languages [3] [15] [4].

Companies must be able to select mitigation strategies that are appropriate for their data, resources, and priorities; however, there is a lack of consensus on how to define and measure bias, making it difficult to compare mitigation strategies across publications. Dataset features like sample size and balance may affect how the bias can be mitigated, making it difficult to extrapolate results from toy problems. Even if the most effective methods are identified, implementing them may require technical expertise, resources, or performance trade-offs some companies cannot afford.

In this paper, we provide a streamlined evaluation of ten gender bias mitigation strategies covering a range of complexity and resource requirements. We evaluate each strategy's ability to mitigate gender bias on the realistic scenario in which a company is using the common language model, RoBERTa, on professional bios to determine subject occupation for the purpose of either recommending job applications or filtering job applicants. Because we use consistent metrics across all strategies and evaluate on the same dataset, we are able to provide a direct comparison of the efficacy of each approach, and we can observe how each strategy is affected by the varying data features of each occupation. By providing clear comparisons of multiple approaches, we hope to make bias mitigation easier to understand and implement appropriately.

Our major contributions are as follows. We demonstrate the presence of gender bias in the outputs of a RoBERTa model trained on the realistic problem of classifying occupation from short bios. We evaluate and compare seven stand alone bias mitigation strategies and three combination strategies in terms of bias mitigation and performance trade-offs. We provide guidance on when to use each approach with consideration to mitigation effectiveness, performance trade-offs, available resources, and data features, such as number of samples and gender imbalance.

2 RELATED WORK

As a baseline for comparison, we re-implement the work done in *Bias in Bios* [2] using RoBERTa [9] instead of LSTMs as well as different evaluation metrics. *Bias in Bios* was a case study demonstrating the existence of gender bias in the high-stakes scenario of occupation classification. The authors quantified gender bias by computing the true positive rate (TPR) gender gap for each occupation, which is defined as the difference in accuracy for each occupation between males and females. They demonstrate that as the percentage of females in an industry increases, the TPR gender gap goes from negative to positive. They also show that removing explicit gender indicators reduces the TPR gender gap but does not eliminate it.

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Measures of intrinsic bias [6] [10] [1], are often used to evaluate models since they are independent of downstream task. However, according to [3] there does not exist an obvious relationship between intrinsic measure of bias and model outputs which are what ultimately impact society. Extrinsic measures of bias must be tailored to the type of task; works that study bias in natural language output or in masked word prediction, such as [8], [14], or Fairscore from [11], do not translate well to bias in classification tasks. Another common approach is to measure gender bias in pronoun resolution tasks, which are hard to generalize. A model may predict male pronouns for a surgeon but correctly predict that the sentence is about a surgeon regardless of pronouns. We choose to measure bias in a real world classification task using the methods described in [12]. The authors present a fairness tree for determining the appropriate metrics for a given scenario. Many of these metrics involve disparity, which is a ratio-based bias evaluation term with general guidelines for determining whether disparity is significant.

3 METHODOLOGY

3.1 Data

The dataset selected for our experiments is the BiasBios dataset from [2]. The text samples are similar to LinkedIn bios, and are labeled with an occupation and gender. The objective is to train a model to fairly predict occupation. To create the original dataset, [2] identified 28 occupations and then scraped bios from Common Crawl using the filtering pattern: "name or pronoun is a ... <occupation>". They removed the matched pattern from the final text. They extracted gender using names and pronouns, and limited labels to "Male" and "Female". We used their code (found at https://github.com/microsoft/biosbias) to generate our base dataset. We then cleaned the data of several systematic inconsistencies associated with certain occupations. Notably, we removed all professors from our dataset as professors often had multiple professions but only one label. The cleaned data consisted of 61,292 training samples and 13,134 validation and test samples.

3.2 Evaluation Metrics

Bias Evaluation. We measure bias using the false negative rate disparity (FNRD) described in [12] with one-vs-rest evaluation for the multi-class case. The FNRD for a given occupation is calculated as the ratio of FNR between a minority gender and a majority gender for that occupation. Since disparity is a ratio, it is assymetric around 1. When plotting, we use the log_2 of the disparity which is symmetric around 0, so that magnitude can be directly compared whether bias favors the majority or minority. The FNR disparity has the following rules for significance defined in [12]:

$$Bias = \begin{cases} \text{significant majority favoring} & \text{if } FNRDisparity_{occ} > 1.25 \\ \text{majority favoring} & \text{if } FNRDisparity_{occ} > 1 \\ \text{perfectly equal} & \text{if } FNRDisparity_{occ} == 1 \\ \text{minority favoring} & \text{if } FNRDisparity_{occ} < 1 \\ \text{significant minority favoring} & \text{if } FNRDisparity_{occ} < 0.8 \end{cases} \quad (1)$$

We report the number of classes for which bias exceeds the significance threshold and the cumulative disparity, which we define as the square root of the sum of squares of the log_2 of class disparities. Plots of per-class disparity can be viewed in Appendix A.

Performance Evaluation. Disparity measures relative performance, so a random classifier would have no disparity. Since in many cases, this would cause more harm for the disadvantaged gender, we also measure absolute performance. We define the *disadvantaged FNR* for each occupation as the false negative rate for whichever gender the baseline performed worst on for that occupation. We also report overall model accuracy on the occupation classification task and the average disadvantaged FNR across classes.

4 EXPERIMENTS AND RESULTS

4.1 Baseline Bias

A real world company wanting to automate text-based decisions is likely to use smaller off-the-shelf classification models finetuned on a unbalanced corpus of available data. As such, for our baseline, we finetuned a RoBERTa Base [9] model on the BiasBios training set by training end-to-end. All hyperparameters are listed in Appendix D. We evaluated our model using the methods described in section 3.2; cumulative results are reported in Table 1, and per-class results are plotted in Figure 1. The baseline model performed well on the classification task with a total accuracy of 85.24%. In terms of bias, 20 of the 27 occupations demonstrated disparity above the significance threshold, mostly favoring the majority gender.

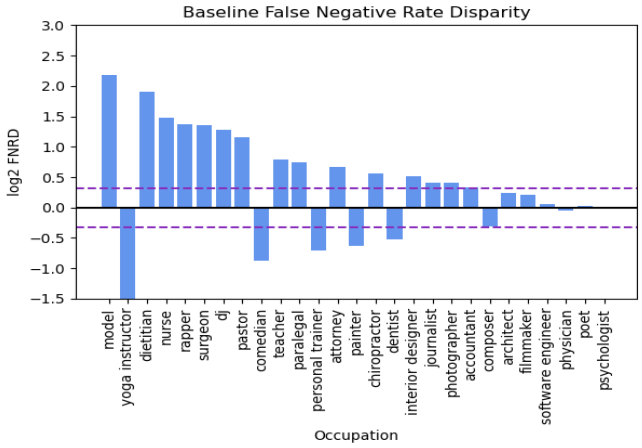

Figure 1: Baseline results for per-class log_2 FNRD defined as the FNR ratio between the majority and minority gender for each occupation. The log_2 is used such that the magnitude of majority and minority favoring disparities can be directly compared. A positive value favors the majority gender while a negative value favors the minority. A perfectly unbiased model has a score of 0, and the dashed lines show the cutoffs for significance.

4.2 Experiments

We implement ten strategies for reducing gender bias and compare them to this baseline. We begin with naive approaches followed by

more advanced methods requiring custom models or training architectures; we end exploring the impacts of combining approaches.

4.2.1 Naive approaches. All naive approaches require minimal additional time, compute, and technical expertise to implement.

Balancing: Our first two approaches involve simply balancing the training data such that each occupation has equal gender representation. We experiment both with down-sampling the majority gender without replacement and up-sampling the minority gender with replacement. We then finetune a RoBERTa-Base model for each balanced dataset and evaluate on an unaltered test-set.

Redacted: For our third strategy, we simply redact all pronouns and names using simple pattern matching approaches similar to [2]. We then finetune a RoBERTa model on the redacted training data and evaluate on redacted test data.

4.2.2 Advanced Approaches. Our advanced approaches involve customization of models, training pipelines, or data generation.

FairBERTa: We experiment with the possibility that a fairer base-model may result in fairer downstream performance. We download the weights for Meta's ResponsibleNLP model FairBERTa, [11], which is pretrained on text with altered demographic patterns. We evaluate the fairness of FairBERTa's downstream decisions by finetuning it on the BiasBios training set and evaluating on the test set. While it uses a custom model, this approach takes no more time, compute, or expertise than the naive approaches.

Fairtuning: In the same publication as FairBERTa, the authors demonstrate that models finetuned on text with altered demographic patterns are less intrinsically biased than those finetuned on natural data [11]. We replicate their perturbation model for swapping explicit gender indicators and apply it to 100% of the samples in our BiasBios training dataset. We then finetune a RoBERTa-base model on the perturbed samples and evaluate on our unperturbed test set. Fairtuning requires technical expertise to write training scripts for the perturbator and takes approximately 24 hours to train and 72 hours to apply on four NVIDIA T4 GPUs.

Adversarial: Instead of focusing on explicit gender indicators, we experiment with removing implicit gender indicators using the adversarial approach described in [7]. We use a RoBERTa Base as our encoder with one MLP head for occupation classification and one MLP head for gender classification. The occupation classifier backpropagates the loss as normal while a gradient reversal layer [5] is added before the gender classification head. Thus, the encoder is trained to create embeddings containing information relevant to occupation classification and to omit information relevant to gender classification. After adversarial training, the base model weights were frozen and used as the base for a new occupation classifying head trained on the Bias Bios training set and evaluated on the test set. Adversarial bias mitigation requires the most technical expertise to implement, but only twice as much time and compute resources as the baseline.

Generative Augmentation: In [13], the authors survey numerous ways to augment training data including generating new data with auto-regressive LLMs. We explore this avenue of data augmentation using a Vicuna 7B model. For each occupation, we identified the disadvantaged gender and generate new samples for that occupation and gender combination. We finetune a RoBERTa base model on the augmented training set and evaluate on the standard test set. Details for prompts and augmentation methodology are in Appendix E. Generative augmentation takes five days on four NVIDIA T4 GPUs to generate the data as well as time and technical expertise to implement and perform prompt engineering.

4.2.3 Combined Approaches. The resources necessary for combined approaches approximately equaled the sum of the resources necessary for each component approach.

Balanced (down-sampled) + Redacted: Balancing the training data is intended to reduce correlations between gender and occupation learned from the training data. Redacting gender terms is intended to reduce model dependency on gendered associations learned in pretraining and fine-tuning. We balance the redacted training data by down-sampling and evaluate on redacted test data to see if the combined approach was more effective than the individual approaches.

Adversarial + Redacted: Redacting names and pronouns scrapes the data of explicit gender indicators, and adversarial training is intended to reduce the influence of intrinsic gender indicators. By combining these approaches, we may be able to reduce more of the gender influence in the decision, and thus more of the bias. We use the redacted training data for adversarial training, and then again for finetuning. We then use the redacted test set for evaluation.

Class-Based Mitigation: For classification problems with multiple classes, the data features of each class may impact the effectiveness of a given approach in different ways. Based on experimentation, we developed a general procedure for class-based mitigation described in Algorithm 1 in which data alterations are made to each class individually. We believe that the reasoning behind each step is general enough to apply more broadly to other datasets, but acknowledge we have not formally tested this assumption. We apply these alterations to the training data, finetune RoBERTa-Base on the altered data, and evaluate on the unaltered test set.

Algorithm 1: Mitigation Strategies

```
if no significant bias for the class in baseline then
    Do not apply any mitigation strategies;
else if bias favors the minority gender in baseline then
    Use generative AI to augment samples for that class;
else if more than 300 training samples and majority gender
    makes up less than 70% of the data then
    Balance the class through down-sampling;
else
    Apply fairtuning to the class;
```

4.3 Results

In Table 1, we report the results for the cumulative metrics described in Section 3.2. Plots of the per-class FNRD for each approach can be found in Figure 2 in Appendix A. All approaches improved cumulative disparity over baseline; however, the degree of improvement and the effect on absolute performance varied by both strategy and class. No approach eliminated significant bias across all classes.

Table 1: Cumulative Results

Experiment	Number w Sig. Bias ↓	Cumulative Disparity ↓	Total-Accuracy ↑	Average Disadvantaged FNR ↓
Baseline	*20*	*5.10*	*85.24*	*26.04*
Balanced (down-sampled)	14	3.59	83.53	27.16
Balanced (up-sampled)	16	4.60	84.57	24.81
Redacted	15	4.23	84.89	23.29
FairBERTa	15	4.67	**84.98**	25.06
FairTuning	15	4.17	84.43	**22.54**
Adversarial	**12**	4.15	77.75	35.77
Generative Augmented	16	4.38	84.92	23.04
Balanced Redacted	17	3.29	83.39	26.59
Redacted Adversarial	**12**	3.44	78.03	32.53
Class-Based Mitigation	**12**	**2.37**	83.14	22.76

4.3.1 Naive Approaches. Balancing the data tended to reduce disparity but increase error for the disadvantaged gender, particularly if the disadvantaged gender was the majority. Balancing through up-sampling was one of the least effective approaches, but down-sampling was fairly competitive. Redacting names and pronouns had middling impact on disparity, but maintained strong accuracy and improved the disadvantaged FNR competitively.

4.3.2 Advanced Approaches. FairBERTa had the smallest improvement in cumulative disparity of all strategies tested, but it also had the smallest negative impact on overall model accuracy and improved the average disadvantaged FNR. Fairtuning had the most positive impact on average disadvantaged FNR as well as a small decrease in overall accuracy. It also reduced disparity for all classes showing significant bias in favor of the majority gender at baseline, but tended to increase disparity if class bias favored the majority gender or there was no significant class bias. Adversarial training had the fewest classes with significant bias, but also showed a drastic decrease in performance. Generative augmentation reduced both disparity and average disadvantaged FNR, and notably was effective when the disadvantaged gender was the majority.

4.3.3 Combined Approaches. Balancing redacted data through down-sampling reduced cumulative disparity more than any mitigation strategy did on its own; however, it had the most classes with significant bias. While the disadvantaged FNR increased compared to baseline, it did not increase by as much as down-sampling alone. The adversarial approach applied to redacted data was largely more effective than adversarial training on unaltered data, as the cumulative disparity decreased and the performance was not as negatively affected. Class-Based Mitigation was by far the most most successful approach, resulting in the largest decrease in disparity and the second largest decrease in average disadvantaged FNR. Adding generative data for cases in which bias favored the minority gender had mixed results, so this step may not be necessary if time and computational resources are limited.

5 DISCUSSION

Organizations often lack the resources, expertise, and time to implement custom AI training, and may resist performance sacrifices from bias mitigation strategies that vary in effectiveness across datasets with different properties. When selecting a bias mitigation approach, an organization should restrict their options based on their constraints, then consider the important features of their data and choose the optimal approach based on how much of a performance trade-off they're willing to accept. In Table 2, we provide guidelines based on our results for selecting an appropriate approach with respect to different constraints including resource limitation, performance trade-offs, and data features.

Resource Constraints. Industry leaders may be able to afford expensive compute and expertise, but companies new to AI may require solutions that use fewer resources. Fortunately, our results demonstrate the naive approaches and FairBERTa can competitively improve bias over baseline at trivial cost and additional expertise. Of the more advanced methods, adversarial training adds the least compute resources, but the most expertise to implement; combination approaches required approximately the summed resources of their components. The remaining advanced strategies require custom machine learning engineering and require multiple days of use on four NVIDIA T4 GPUs, requiring higher monetary investment. Additionally, if model updates need to happen often, strategies that take days of additional time may become impractical, making these approaches expensive from a time perspective as well.

Data Features. Performance for different mitigation strategies varied across occupations based on per-class data features, particularly number of training samples available for the disadvantaged class and whether or not baseline bias favored the majority class (see Figure 2 in appendix A). When there are few samples, the variance for the adversarial approach increases, and balancing through down-sampling also hurts performance when there are fewer samples in the minority gender. Furthermore, balancing approaches and fairtuning led to increased disparity for classes where the disadvantaged gender was the majority. FairBERTa, the adversarial approach, and generative augmentation were all effective in this case, as were any combined approaches including them.

Performance. If reducing disparity is the sole objective, the most effective approaches are class-based mitigation, balancing through down-sampling with redaction, and adversarial training on redacted

Table 2: When to Use Each Strategy

Mitigation Technique	Limited Compute	Limited Time	Limited Expertise	Few Training Samples	Bias Favors Minority	Minimize Disparity	Minimize Performance Trade-offs
Balanced down-sampled	✔	✔	✔	×	×	✔	×
Balanced up-sampled	✔	✔	✔	✔	neutral	×	×
Redacted	✔	✔	✔	✔	neutral	×	✔
FairBERTa	✔	✔	✔	✔	✔	×	✔
FairTuning	×	×	×	✔	×	✔	✔
Adversarial	neutral	neutral	×	neutral	✔	✔	×
Generative Augmented	×	×	×	✔	✔	×	✔
Balanced Redacted	✔	✔	✔	×	×	✔	×
Redacted Adversarial	neutral	neutral	×	neutral	✔	✔	×
Class-Based Mitigation	×	×	×	✔	✔	✔	✔

data. We advise avoiding using FairBERTa, balancing through upsampling, and generative augmentation. Balancing through downsampling was very effective at decreasing cumulative disparity, and combining it with redaction resulted in one of the lowest cumulative disparities, but the highest number of classes with significant bias. If minimizing performance trade-offs is important, the landscape of options changes. The adverse performance effects of adversarial approaches are likely too great for it to be worth considering in most circumstances, and balancing through downsampling may reduce absolute performance for the disadvantaged gender. The most effective methods for improving absolute performance for the disadvantaged class were fairtuning, the class-based approach, generative augmentation, and redacting gendered terms.

6 CONCLUSION

As the impacts of AI move into commercial deployment, companies need to be able to directly compare the benefits and trade-offs of bias mitigation strategies and make informed decisions on which approaches are most appropriate for their data and resources. We demonstrate the presence of bias on a realistic baseline and provide an evaluation of ten bias mitigation strategies of varying complexity, as well as the means to compare the efficacy of future strategies. Our results show that selecting the best approach depends upon the constraints and data features of the scenario at hand and that naive approaches can outperform more complex strategies. We recommend organizations use the guidelines we provide to narrow their options based on their constraints, consider the features of their data, and select the best approach based on their metric priorities.

7 FUTURE WORK AND LIMITATIONS

This work is limited in that no approach effectively eliminates bias, there is limited explainability on why each approach is effective, results are only collected for one dataset, and only a small set of strategies are considered. In the future, we would like to identify another quality dataset on which to test each of these approaches and confirm the results described here generalize well. Additionally, we could identify any commercialized tools and more recent, state of the art approaches for bias mitigation and use our framework to evaluate their effectiveness. Finally, we can apply explainability tactics to better understand why each approach was effective.

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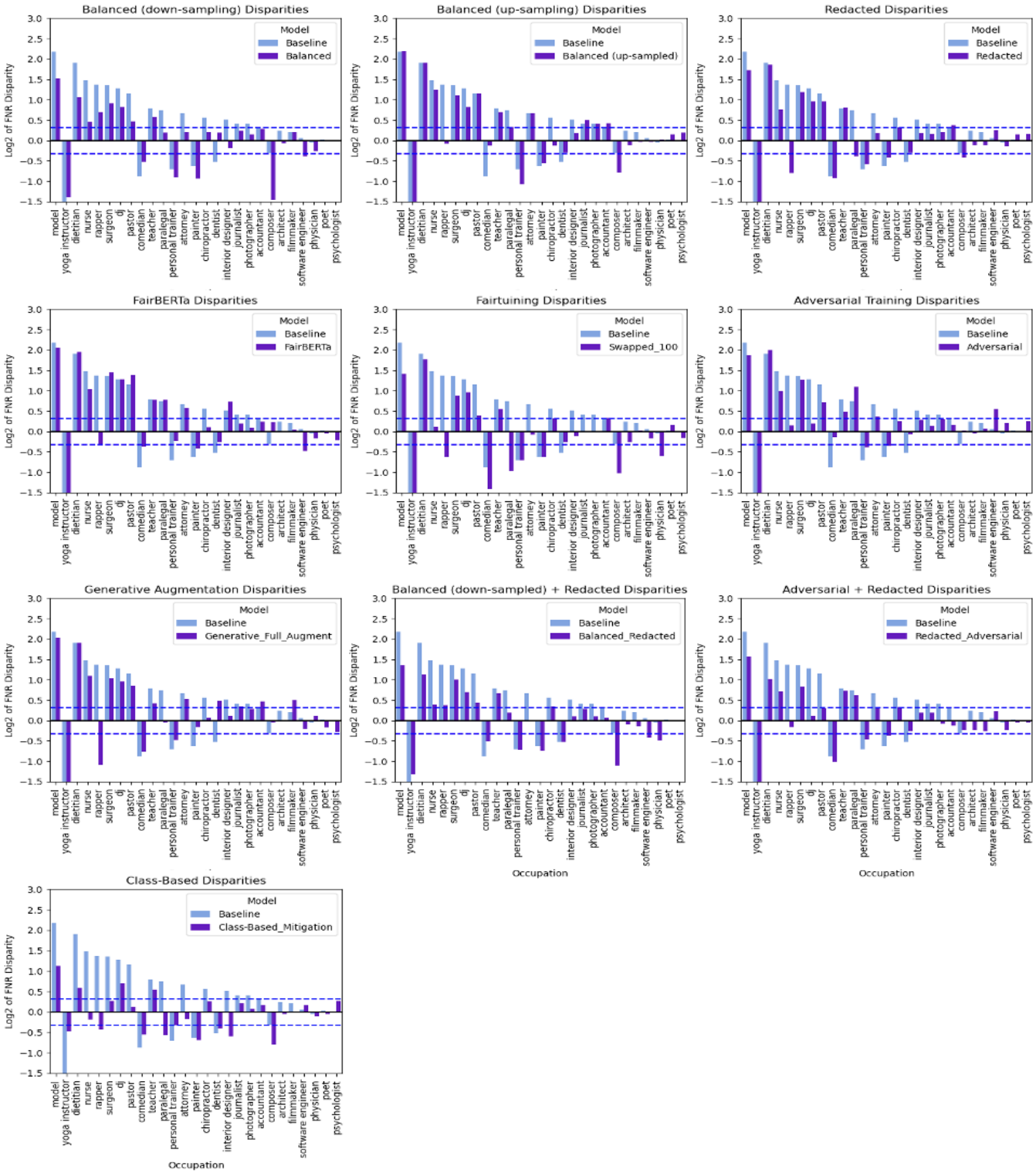

Figure 2: Results for per-class false negative rate disparity for all mitigation strategies. The log_2 of the false negative rate disparity between the majority and minority gender for each occupation. The log_2 is used such that the magnitude of majority-and-minority favoring disparities can be directly compared. A positive result favors the majority while a negative result favors the minority. Zero is the score for a perfectly unbiased model, and the dotted lines show the cutoffs for significance. The occupations are ordered from greatest to least disparity in the baseline case.

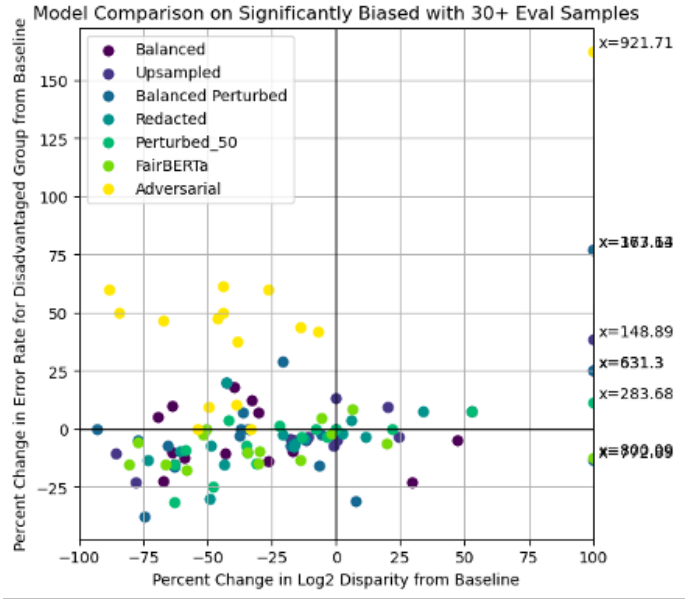

Figure 3: This figure plots percent change in the log of the false negative rate disparity against the percent change in the error rate for the disadvantaged group from baseline for each mitigation strategy (separated by color) and each occupation for which there was significant bias in the baseline or the mitigated results. The top right quadrant indicates an increase in disparity and error rate making this the worst quadrant. The bottom left quadrant indicates a decrease in both disparity and error rate. Points in the top left demonstrated a decrease in disparity but resulted in worse absolute performance for the disadvantaged class compared to baseline. Points in the bottom right demonstrated an increase in disparity, but improved absolute performance for the disadvantaged class compared to baseline.

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A PER-OCCUPATION RESULTS

In Table 3, we list the data statistics for each occupation in the dataset used for training the baseline. In Figure 2, we directly compare each mitigation strategy to baseline by plotting the log of the false negative rate disparity for each occupation. This breakdown demonstrates the effectiveness of each strategy on individual classes with different characteristics. Each approach decreases disparity for some categories and increases it for others. Figure 2 in conjunction with Table 3 can be used to assess the effectiveness of different strategies on classes with different data features. For instance, balancing through down-sampling has the greatest impact on classes with the most disparity, many of which also have heavily imbalanced training data.

The performance and disparity trade-offs can be observed in a two dimensional scatter plot in which percent change in disparity from baseline is on the x axis and percent change in error rate from baseline for the disadvantaged group is on the y axis. In Figure 3, for each mitigation strategy, we plot these metrics for individual occupations, allowing us to compare trends in mitigation effectiveness and performance trade-offs. For example, the adversarial model consistently reduced disparity at the cost of absolute performance.

B BALANCE TRENDS

In Figure 4, we compare how the data imbalance and number of training samples influence the percent change in Disadvantaged FNR from baseline for balancing through downsampling and up-sampling. We find that the more imbalanced a class is, the more balancing it by down-sampling increases the error rate for the disadvantaged group compared to baseline. Similarly, the fewer training samples there are for the minority gender in a given occupation, the more the error rate increases. We see the reverse trend with balancing through up-sampling. The large variance for classes with fewer than 300 training samples for the minority gender or more than a 70% data imbalance influenced our design of Algorithm 1 for the combined approach.

C AEQUITAS FAIRNESS TREE

To select our metrics, we used the Aequitas fairness tree as shown in Figure 5 [12]. In this paper, we consider the scenario where a large language model is ingesting bios taken from resumes and outputting the applicant's occupation as part of a filter to determine which applicants are qualified for human review. We suspect that unfairness comes in part from bias in the model itself. We largely trust the labels; since they were automatically generated from a template there is no biased human in the system to bias the labeling process. Since, in this scenario, those correctly classified are more likely to get a good job they're qualified for, we consider

Table 3: Per-Occupation Data Statistics

Occupation	Total in Training Data	Num M in Train Set	Num F in Train Set	Percent-Majority (train)	Num M in Test Set	Num F in Test Set
dietitian	1,116	62	1,054	94.44%	*8*	217
nurse	2,863	207	2,656	92.77%	39	550
rapper	279	246	33	88.17%	57	*11*
paralegal	471	65	406	86.20%	*10*	92
surgeon	2,687	2251	436	83.77%	519	92
model	1,665	274	1,391	83.54%	75	298
software engineer	2,258	1,879	379	83.22%	358	92
yoga instructor	580	103	477	82.24%	*14*	101
interior designer	435	79	356	81.84%	*15*	57
dj	337	270	67	80.12%	30	*6*
composer	1,533	1,210	323	78.93%	280	67
comedian	681	508	173	74.60%	107	49
psychologist	4,386	1,423	2,963	67.56%	312	674
pastor	747	499	248	66.80%	102	49
architect	2,167	1,439	728	66.41%	325	132
chiropractor	568	371	197	65.32%	72	49
teacher	3,823	1,382	2,441	63.85%	287	554
filmmaker	1,989	1,207	782	60.68%	220	169
photographer	6,560	3,864	2,696	58.90%	846	608
dentist	2,231	1,292	939	57.91%	301	216
poet	2,204	939	1,265	57.40%	199	269
journalist	6,865	2,537	3,328	56.69%	1,016	826
attorney	8,365	4,742	3,623	56.69%	1,016	826
accountant	1,346	725	621	53.86%	166	120
painter	2,215	1,057	1,158	52.28%	218	256
physician	3,582	1,867	1,715	52.12%	392	366
personal trainer	339	165	174	51.33%	33	44

the intervention helpful or assistive. As such, we select to use the false negative rate disparity as our metric for identifying bias.

D HYPERPARAMETERS

Our models were all implemented using Huggingface and trained end-to-end using the Trainer API. We began by performing a hyperparameter grid search on our baseline model for learning rate, training batch size, and number of epochs. We used the defaults for all remaining hyperparameters. The optimal values we used were 12 epochs, a training batch size of 8, and a learning rate of 1e-5. We used these hyper parameters for all bias mitigation strategies except for the adversarial approach which was unstable and required another round of hyperparameter tuning. For the adversarial model, we trained for 12 epochs with a batch size of 8 and learning rate of 8.9E-6.

E GENERATIVE AUGMENTATION PROMPTS

We generated data for augmentation using a Vicuna 7B model. For each occupation, we identified the disadvantaged gender and generated new samples for that occupation and gender combination by feeding the model the following prompt:

> Generate 10 short, LinkedIn style bios for <gender> <occ>. Bios should be 1-3 sentences and describe <occ> from a variety of schools, technical backgrounds, and specialties. They should not all follow the same format. Some example bios are:...

followed by a rotating sample of three bios selected from our existing data. We repeated this process with new example bios until we had produced as many augmented samples as were in the advantaged class, or we had produced twice as many samples as were in the disadvantaged class. We did not balance gender representation, since that would not lead to any changes if the bias favored the minority gender. We chose this strategy because it produced nearly balanced classes in when the disadvantaged gender was the minority, and generated no more than twice as many samples when the disadvantaged group was the majority

F RESEARCH ETHICS AND SOCIAL IMPACT

F.1 Ethical Considerations Statement

In conducting our research, we addressed several ethical concerns related to responsible data usage and privacy. Data was obtained

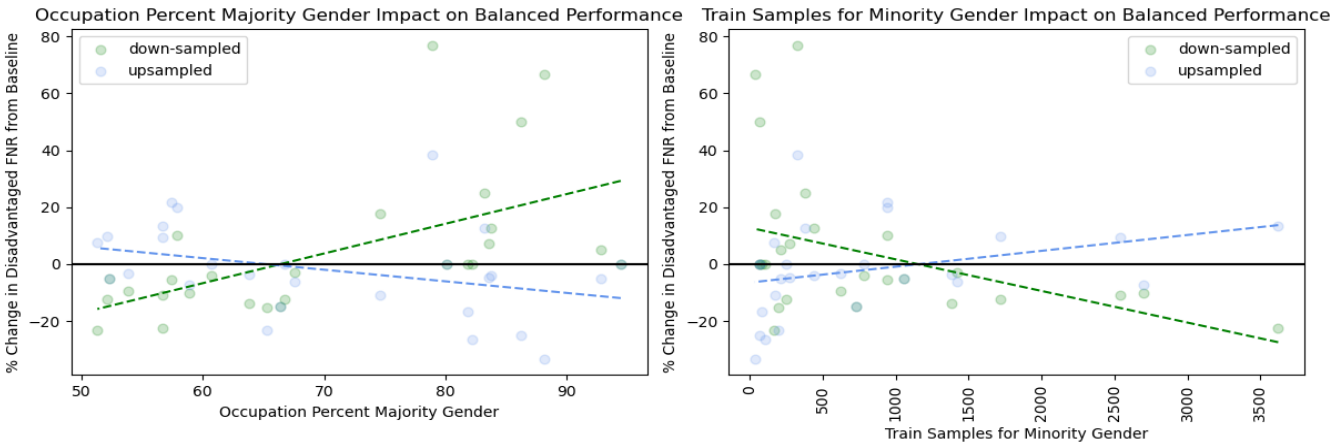

Figure 4: Left: The percent change in the false negative rate for the disadvantaged gender from baseline as a function of the imbalance of the dataset when applying balancing through down-sampling (blue) and upsampling (green) with trend lines. Right: The percent change in the false negative rate for the disadvantaged gender from baseline as a function of the number of training samples for the minority gender when applying balancing through down-sampling (blue) and upsampling (green) with trend lines. Each dot represents a class. Positive values show an increase in error rate for the disadvantaged gender while negative values show improved accuracy.

by scraping short, work-related bios from Common Crawl. Sensitive attributes and labels were inferred from explicit names and pronouns in the text itself. To mitigate privacy risks, we collected only publicly-available information, and stored it on a password-protected, VPN-protected EC2 instance. Moreover, all models trained on the data were locally stored and similarly password-protected. We did not release the data or any models trained on this data, in order to prevent any potential misuse. These actions were taken to uphold the ethical standards of data privacy and security throughout the research process.

F.2 Researcher Positionality Statement

The authors of this paper are a white woman and a black woman working as AI researchers at <redacted for review anonymity>. Our mutual experiences have sensitized us to the importance of addressing bias in technology, particularly to bias against under-represented groups in certain career paths. These viewpoints allow us to comprehensively address bias mitigation in our research. Furthermore, we have experience with how companies implement AI systems and what priorities they may consider when choosing a bias mitigation strategy or choosing whether or not to attempt mitigation in the first place.

F.3 Adverse Impact Statement

This work evaluates several approaches to mitigating bias against a variety of measures that companies might consider when implementing AI systems with the potential for introducing bias. In our paper, we show that some naive and simple approaches can be more impactful than more complex approaches in certain circumstances. A potential adverse effect of this paper could be that companies may be less willing to thoroughly research all of their options for bias mitigation, and instead defer to the most accessible approaches addressed in this paper. Additionally, while all approaches decreased gender disparity, no approach eliminated it. A company using any one of these tactics will still be releasing a model that has disparate errors on the basis of gender. This paper encourages the use of mitigation strategies over no intervention, but that doesn't mean that the AI systems that use them are harmless.

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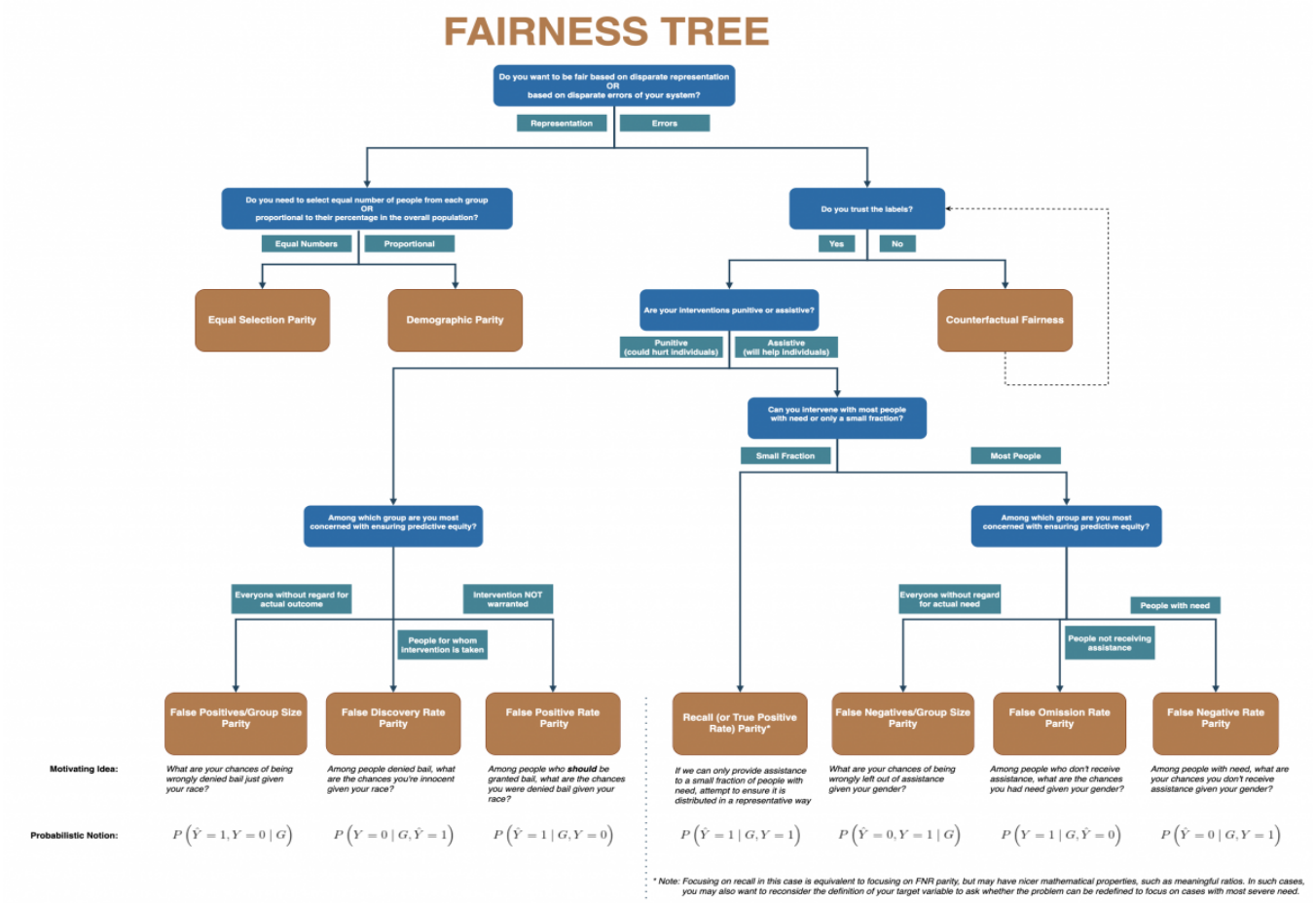

Figure 5: The Aequitas fairness tree from [12] at http://www.datasciencepublicpolicy.org/our-work/tools-guides/aequitas/ used for identifying the appropriate fairness metric for a given dataset and problem.